



Andhra Pradesh State Road Transport Corporation(APSRTC) Visakhapatnam City Bus Transit Analytics -Case Study



EXECUTIVE VIEW

Inference: APSRTC Vizag has recovered fully from COVID with 2.5L daily passengers in FY24, but the EPKM–CPKM gap (₹37.6 vs ₹41.2) means every km operated loses ₹3.6.



BUS FLEET

Inference: BS6 buses - 374 (59% of fleet), BS3 buses -35 (need immediate replacement), 2018 procurement of 118 buses will cross the 15-year threshold by 2033. EV replacement is the optimum solution for these buses



BUS DEPOT

Inference: Maddilapalem needs a dedicated investigation — it has the highest route count (22) and good fleet utilisation (89.1%) but the OTP (82.9%) is less compared to all other depots.



REVENUE

Inference: Routes 38IT,1T has maximum passengers and revenue. whereas routes routes 25M and 5D has good revenue similar in lines to routes 38IT,1T. Increasing number of buses in these routes increases revenue to government as well services to the public

PREPARED BY

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APSRTC Dashboard-Executive View



2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
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635

Total City Buses

84

Active Routes

567

Effective Fleet Operatng

2.5L

Daily_Passengers

2.03L

KM Operated/Day

48.34

EPKM(Rs/Km)

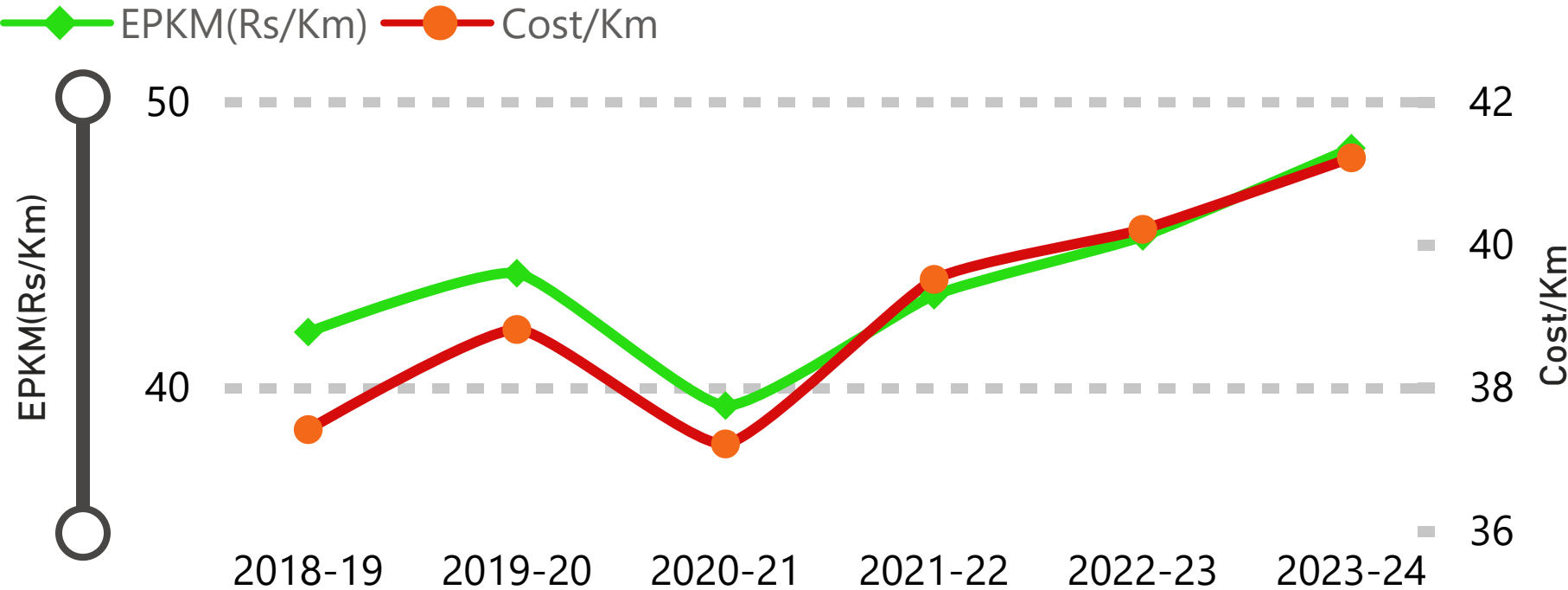
98.0L

Daily Revenue

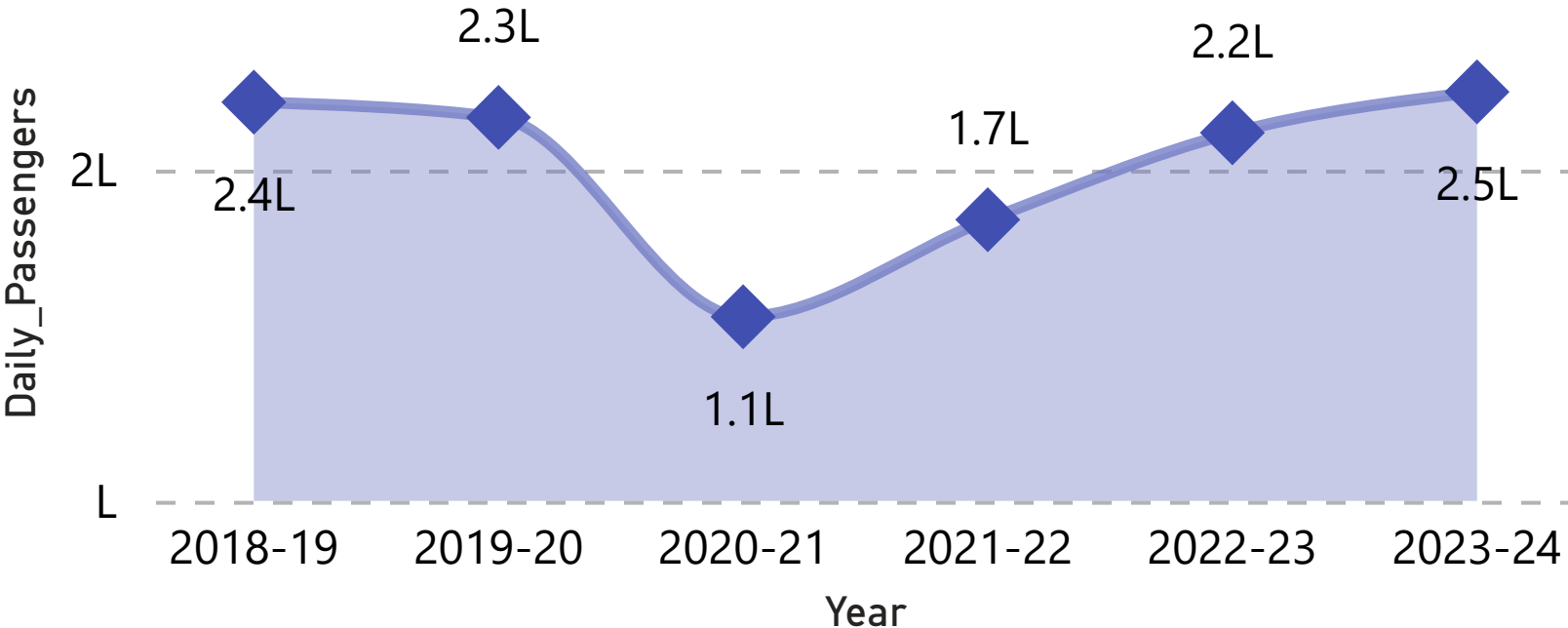
89.29

Fleet Utilization%

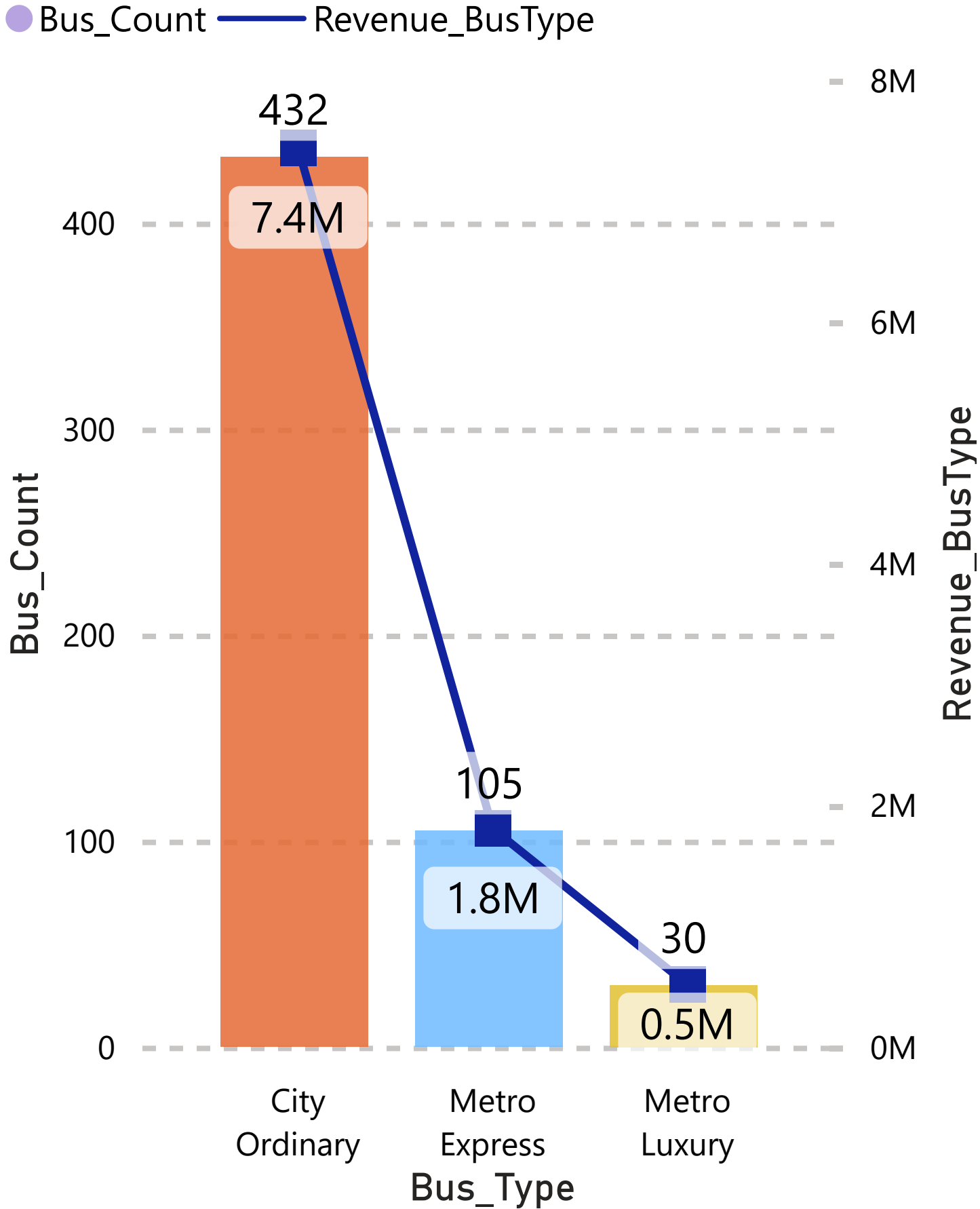
Revenue vs Cost/Km (Rs/Km)



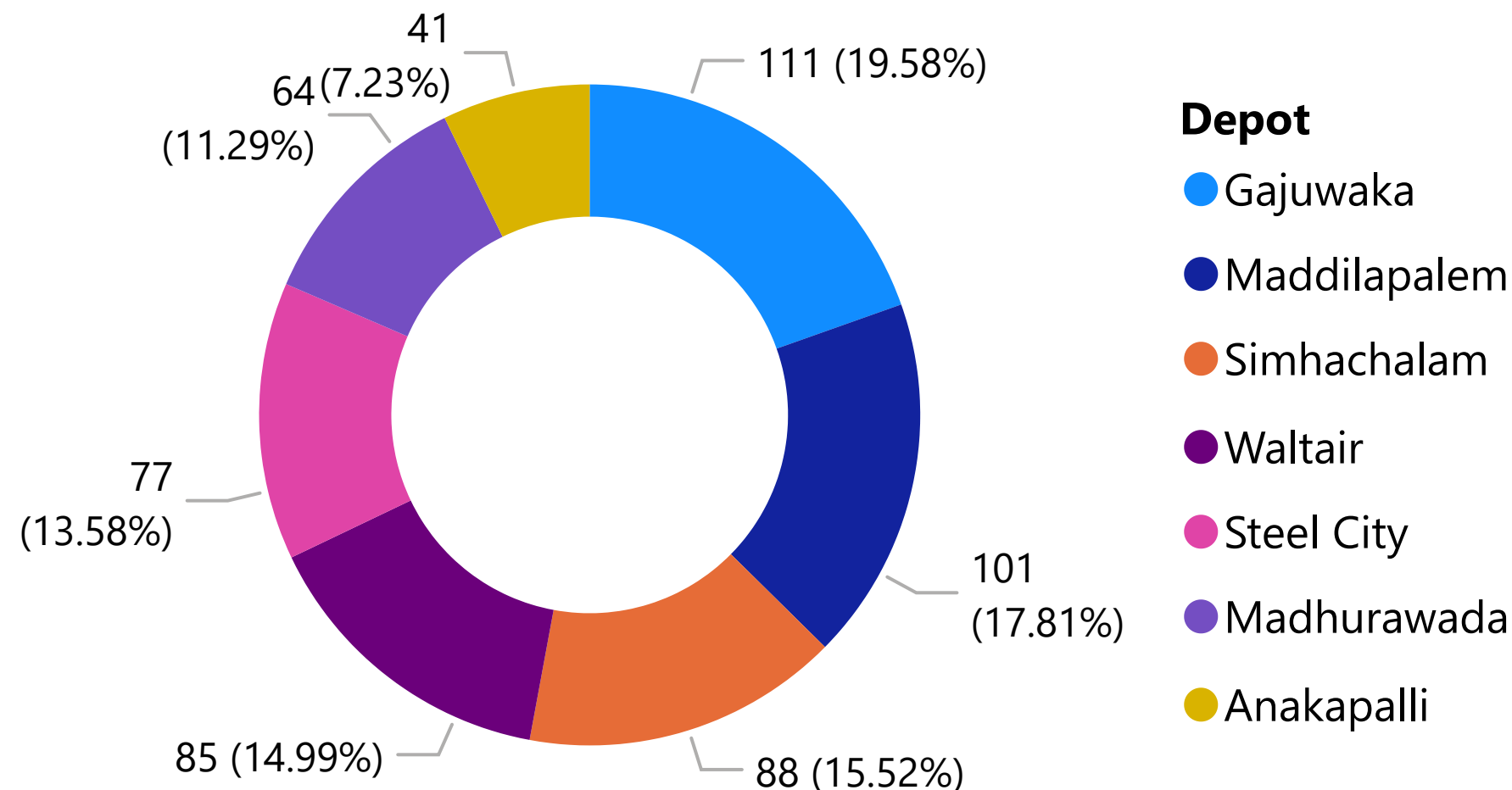
Daily_Passengers Trend (Lakh)



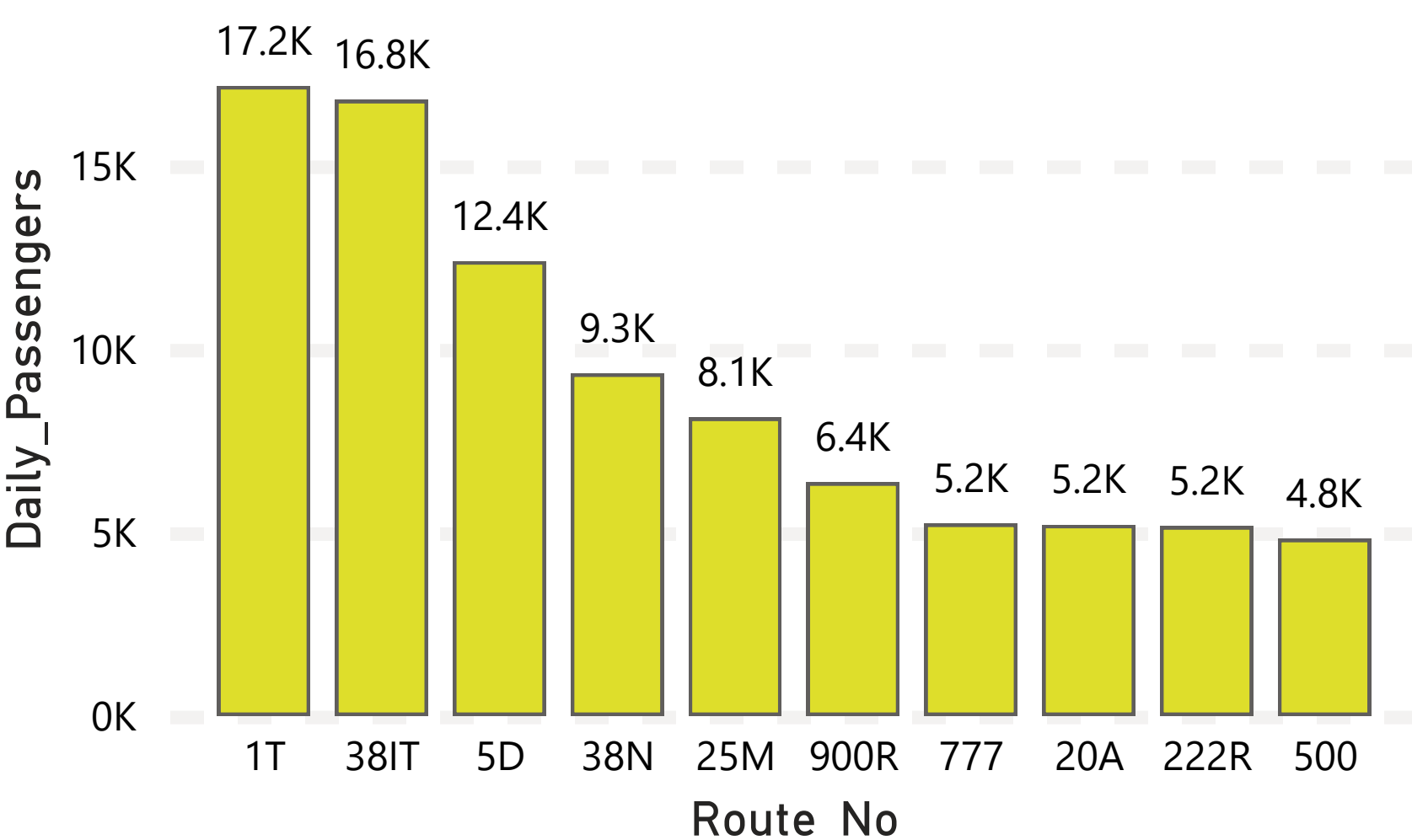
Bus_Type Vs Revenue (Yr:2024)



Fleet by Depot (Yr:2024)



Top 10 Routes-Daily Passengers(Yr:2024)





APSRTC Dashboard- Buses Fleet Analysis



2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
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635

Total City Buses

68

Maint_Buses

17328

Revenue/bus/day(rs)

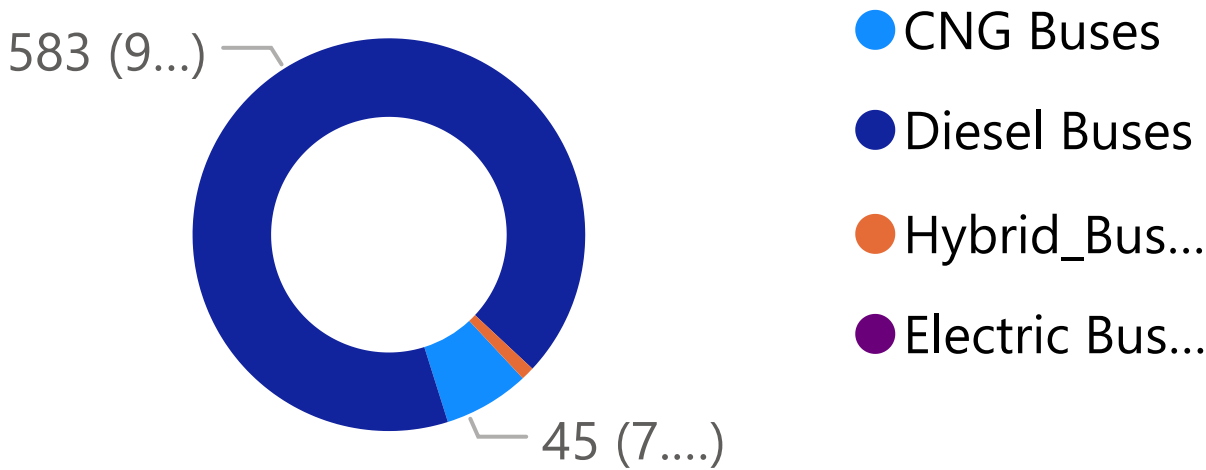
358

Km/Bus/Day
(Avg)

64

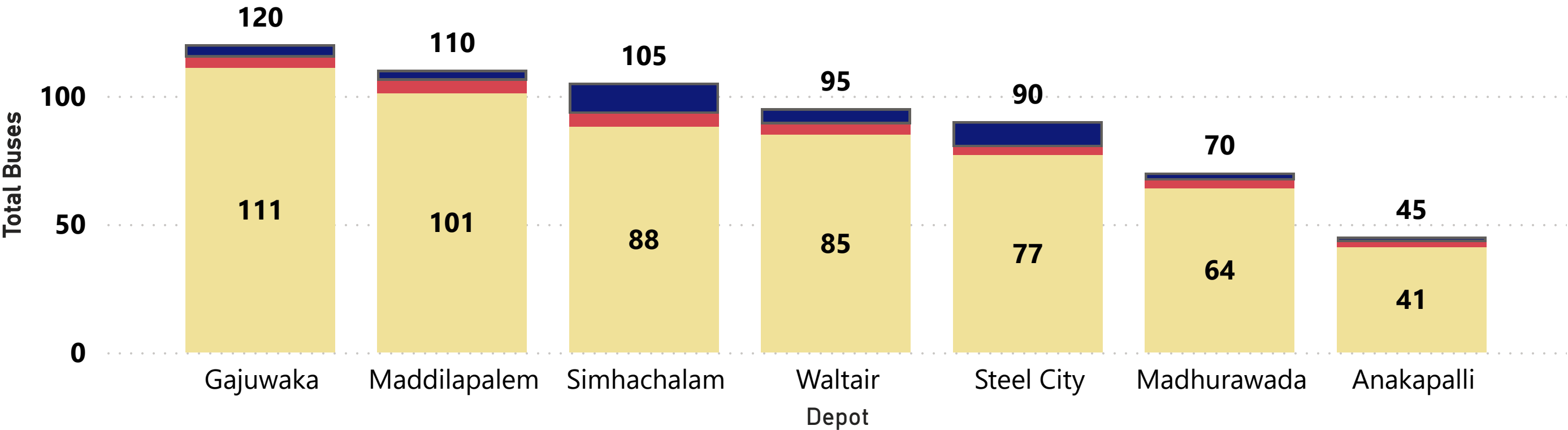
Fleet_age> 10 yrs

Buses_By Fuel Type

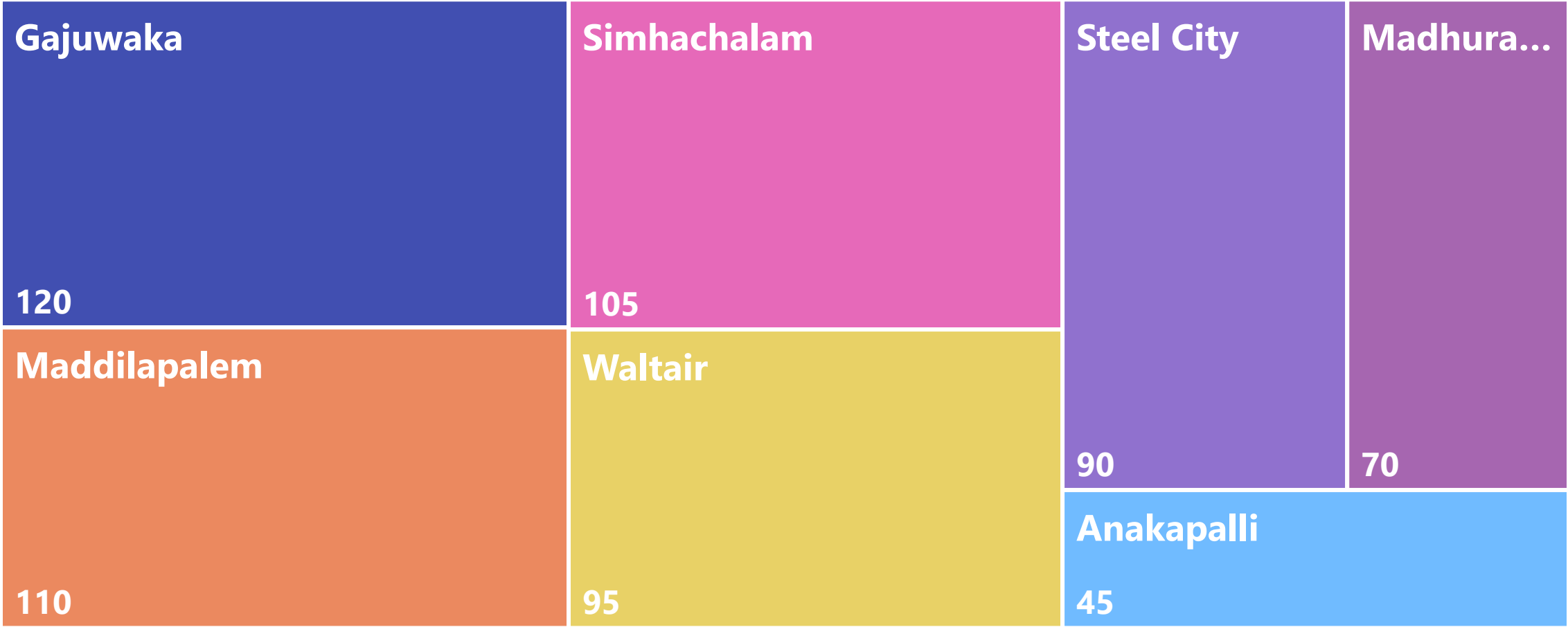


Depot's Buses status (Active/Spare/Workshop,Yr:2024)

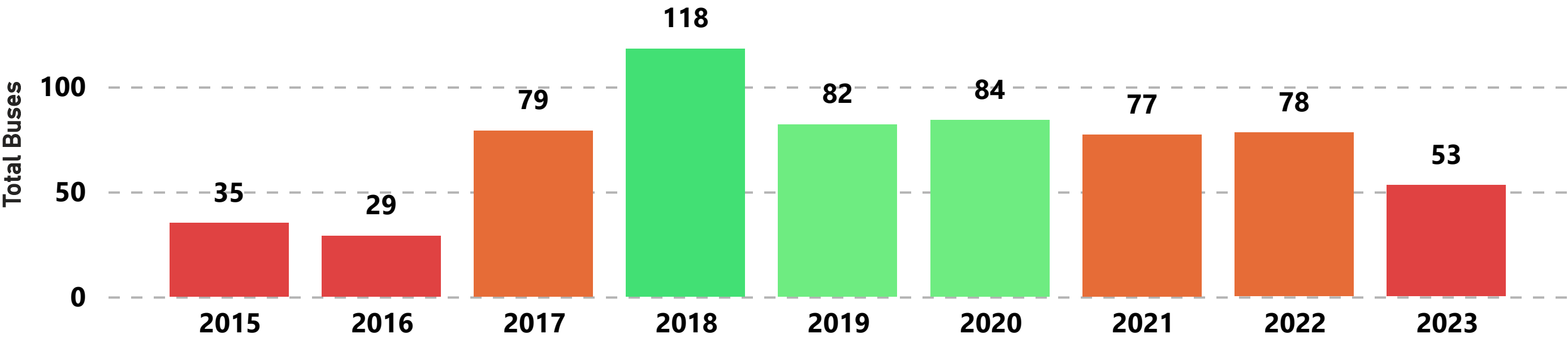
Buses--- Active Spare Workshop



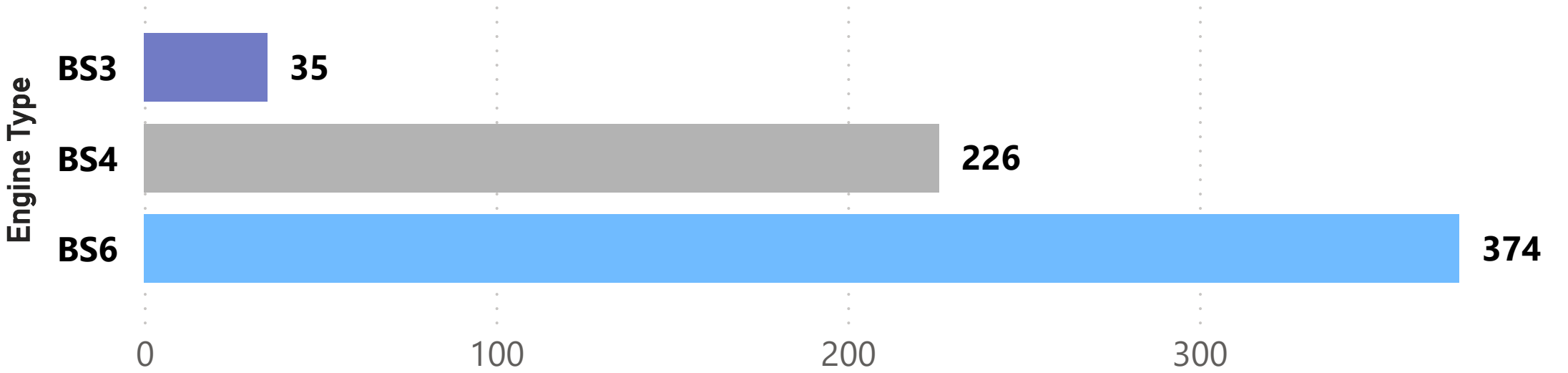
GPS_Fitted Buses/Depot(Yr:2024)



Buses_Manufactured/Year



Buses_by Engine Type and Bus Type,Yr:2024)





APSRTC Dashboard - Depot Score card



2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
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7

Bus Depots

450

Bus Stops / Shelters

41.20

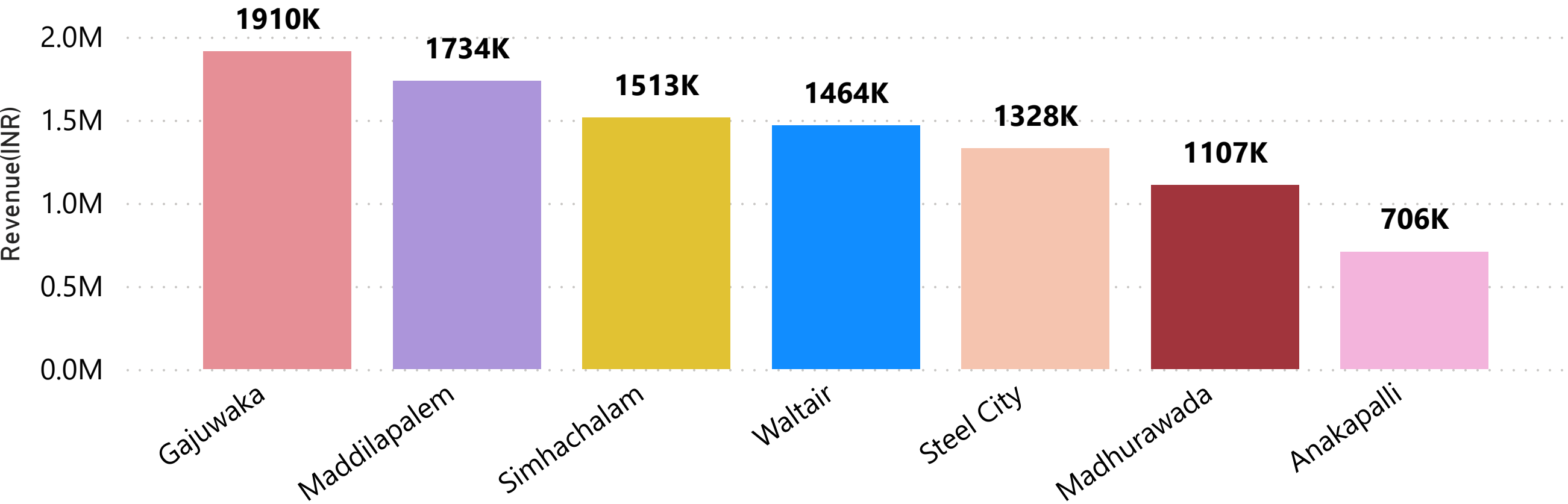
Cost/Km

16.27

Revenue_Growth_YOY

Depot	Buses_count	Total_Passengers	Routes_travel	Total_Revenue	On-Time Performance %
Anakapalli	41	17706	4	706446	91.49
Madhurawada	64	27480	8	1106783	88.40
Steel City	77	32953	7	1328076	91.61
Waltair	85	36450	12	1464258	80.60
Simhachalam	88	37804	20	1513237	87.20
Maddilapalem	101	43546	24	1733957	82.87
Gajuwaka	111	47722	19	1910135	86.51
Total	567	243661	83	9762892	86.16

Revenue/Day_Depots_Wise(Yr:2024)



15.34

EPKM Growth %

Earning/km Growth % is increment in earning/km from 2018-19 to 2023-24

122.12

Pax_Recovery %

Passenger Recovery from 2021-22 to 2023-24 due to Post Covid. Total no. of Passengers Increased Steadily

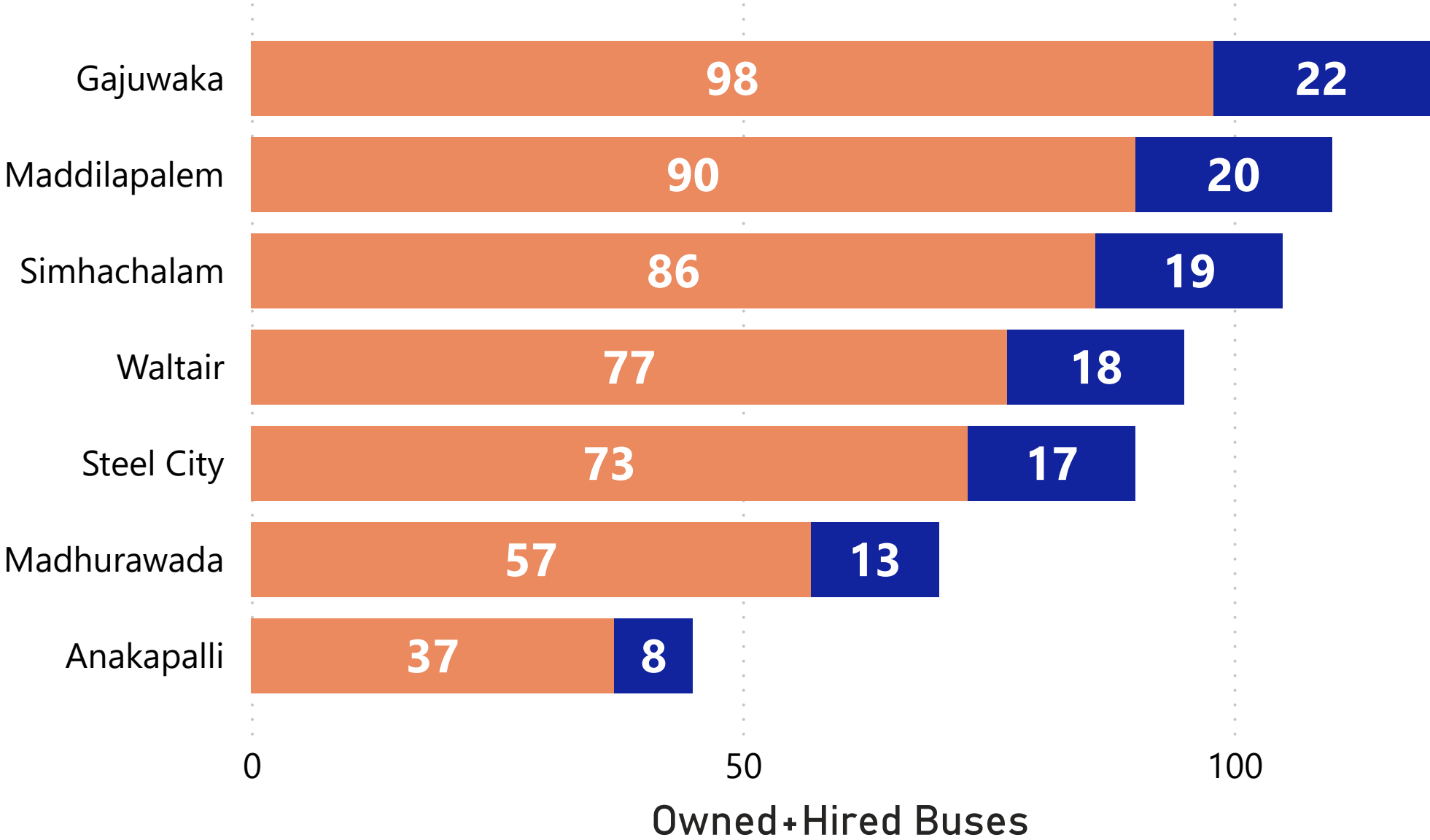
1.07

Fleet CAGR %

Rise in Bus number from 2019-20 to 2023-24. Gradual increase in Bus numbers is observed

Depots vs Buses (Own + Hired) (Yr:2024)

Buses--
Owned_Buses
Hired_Buses





APSRTC Dashboard-Routes, Revenue & Operations



2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
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98

GPS / AIS-140
Fitted Buses(%)

82.00

CO2 Emissions per
Pax-KM (gms)

91.20

ePOS Digital
Ticketing %

4.80

Avg_Trip_Del
ay(min)

15.00

Bus Breakdown
Incidents per...

2.9%

Breakdown
rate

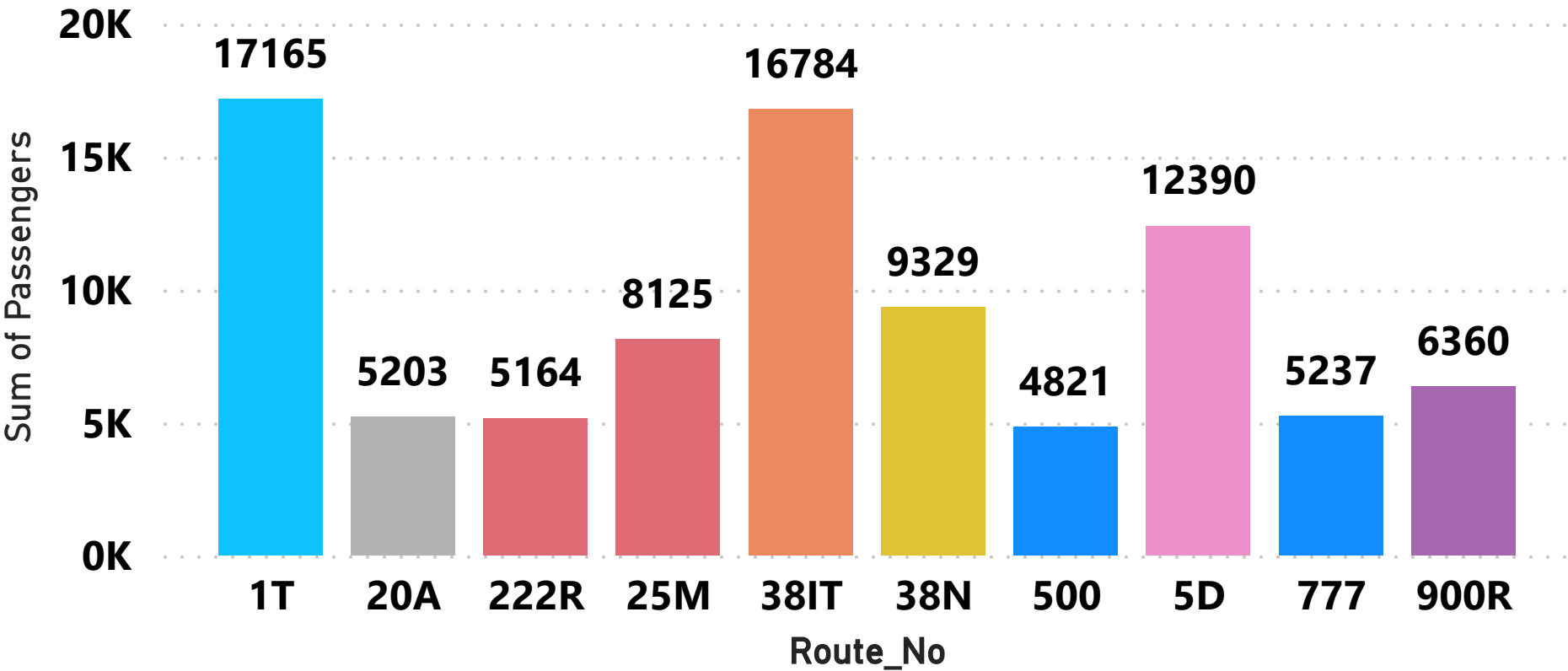
2,384

Total Trips

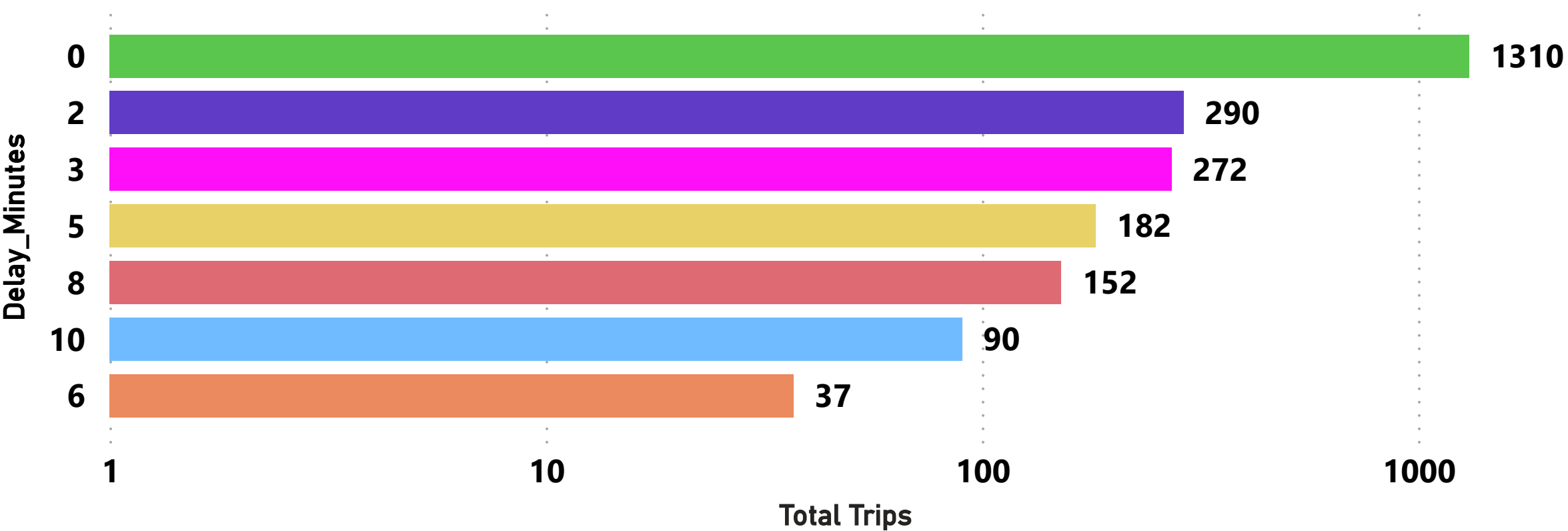
2054

Trips on
Time

Top 10 Routes_Passengers Count



Trips vs Delay_Min



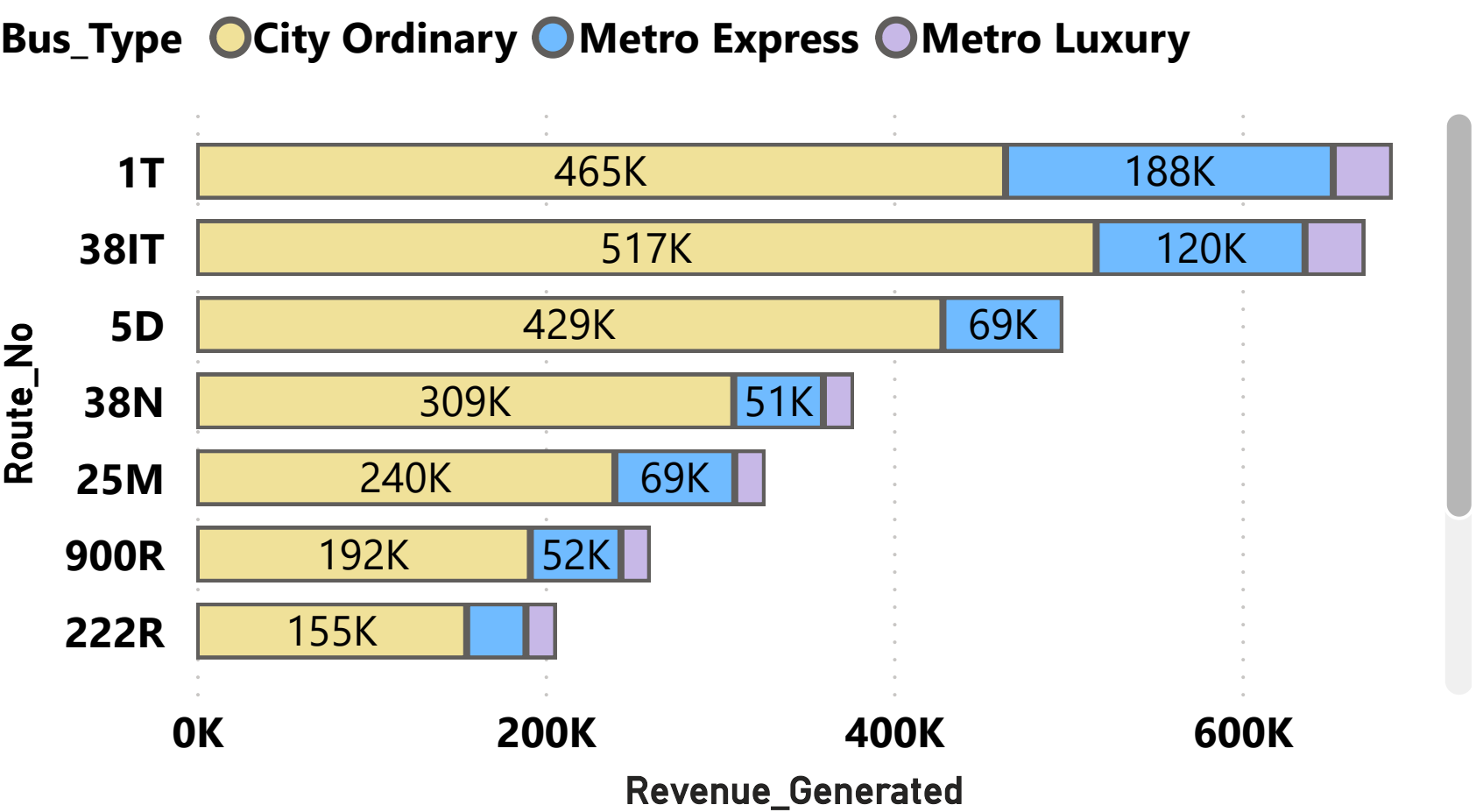
25

Avg_Route_Leng
th(km)

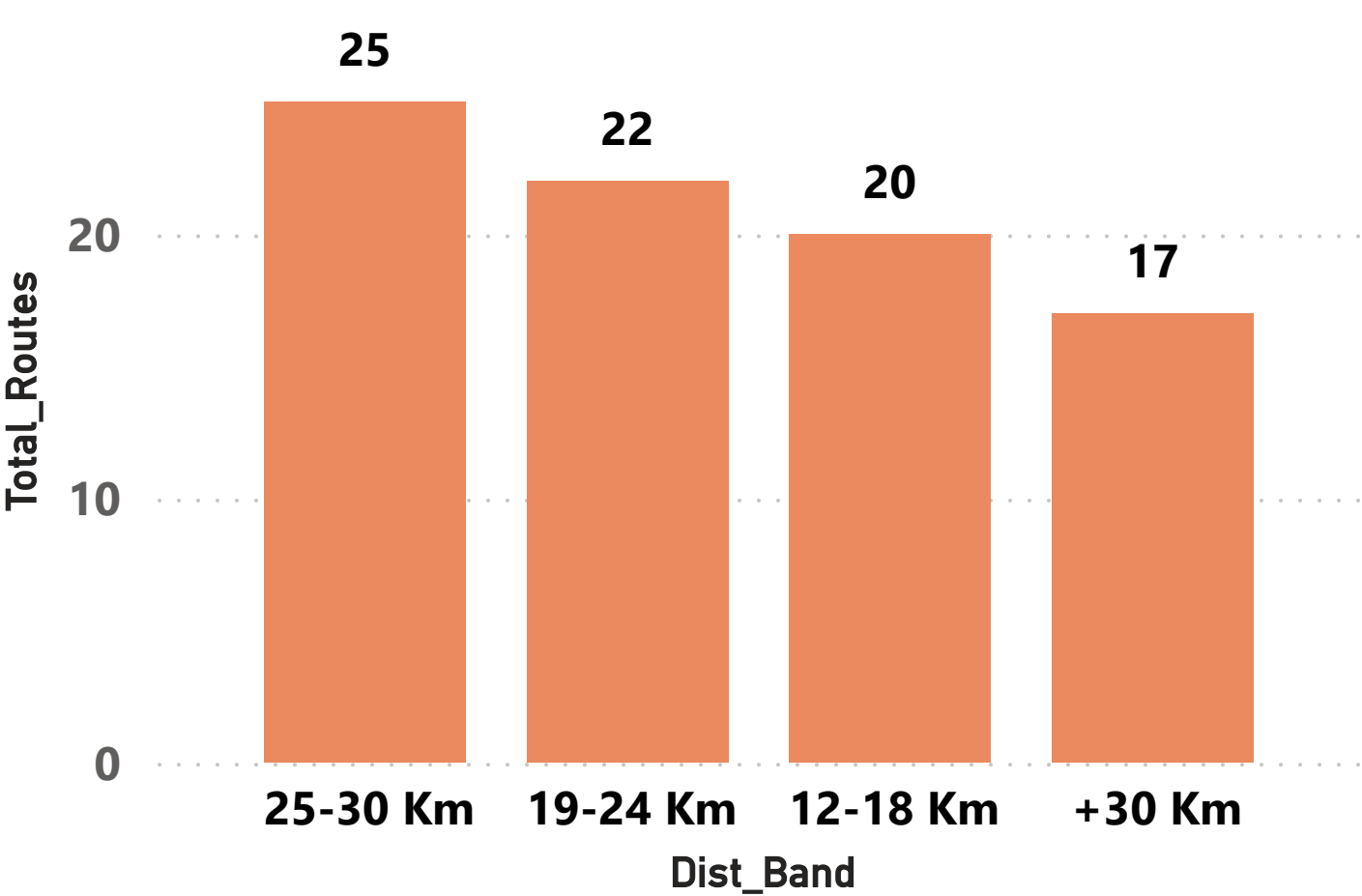
86.16

On-Time
Performance %

Top 10 Routes_Revenue Generation



Routes Coverage vs Distance



Routes Coverage vs Time

